

**Assessment of Societal Issues of the Typhoon Hato in 2017,
and Possible Remedial Measures for the Macau Government**

WONG, Hok Fong

Department of Computer Science and Engineering, The Chinese University of Hong Kong

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Mr. Lau Ka Kiu Jimmy

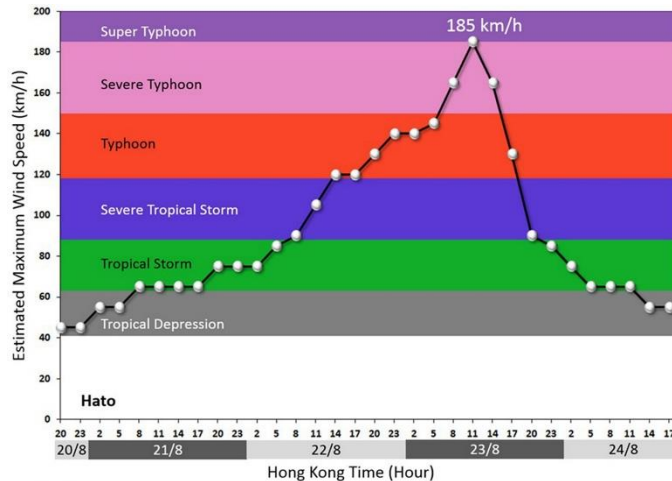
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Introduction

Tropical cyclones are destructive low-pressure storm systems over tropical or subtropical areas, with atmospheric rotation causing heavy rainfalls and windstorms (Lau et al., 2021). Anomalous climatic events, such as El Nino and La Nina, have increased the intensity, frequency, and extent of loss induced by tropical cyclones (Chan et al., 2018; Lau et al., 2021). As such, Macau will be more afflicted with the additional challenges posed. On August 23, 2017, Hato struck the Pearl River Delta subsequent to its landfall after rapid intensification from category-1 to category-3 within 5 hours (Yang et al., 2019). The minimum central pressure was 945 hPa, and the maximum 10-min sustained wind speed was 185 km/h. It is the first time for Macau to issue the No. 10 typhoon signal in 18 years after 1999 Typhoon York (Choy et al., 2022; Lau et al., 2021). Tropical cyclones have posed catastrophic impacts on society and thus are considered crises, which by definition, is the "aggregation of negative scenarios with high uncertainty" requiring immediate solutions (Chan et al., 2018). Based on the stance of crisis management, this essay aims to investigate the aftermath of Hato on a societal level, including psychological impact and financial loss, and combat deficiencies such as optimizations of forecasting procedures and civil defense in the post-disaster response and recovery stage.

Figure 1

Time Series of the Maximum Sustained Wind Speed over the Storms of Hato



Note. From “Super Typhoons Hato and Mangkhut, part I: Analysis of maximum intensity and wind structure,” by C. Choy, D. Lau, & Y. He, (2020), *Weather*, 77(9), 314–320. <https://doi.org/10.1002/wea.3797>

Problems Created and Their Related Solutions

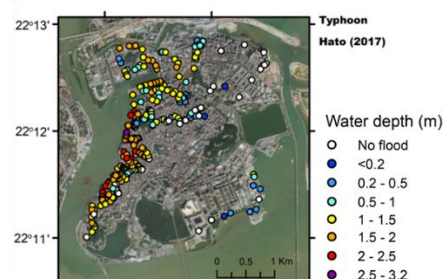
Casualties and Financial Losses

As Chow et al. (2019) have highlighted, the cataclysmal effect of Hato has three main components: acute destructiveness, financial disruption, and fatalities. The uniqueness of the industrial structure in Macau, with foreign-invested gaming licenses providing stable support for the gambling industry and the attractiveness of its heritage and culture contributing to the tourism industry, are sensitive to weather crisis, which could result in a tremendous loss in the economy, urging for the improvement of crisis management from policymakers. Macau Government Tourism Office halted the tour groups until the end of August, creating a strong

negative effect on the two pillar industries. Corroborated by U and So (2020), safety is the top priority for travelers to plan their tours since any variations in tourist confidence will directly affect the GDP of Macau. From a statistical perspective, there were 12 confirmed deaths and 244 injuries in Macau, along with a direct economic loss of USD 1.56 billion (Lau et al., 2021). A field survey has shown the seriousness of coastal inundation by storm surges in inner harbor areas, which reached 2.4 meters with amplification effects of astronomical tides (Yang et al., 2019). Even worse, the complete blackout of the water and electricity supply affected over 20,000 households, which merely reached full recovery after six days (Shan et al., 2018). While Hato was not the most robust tropical cyclone in the last several decades, it is posited that extreme climate change has vastly influenced the features of tropical cyclones in the Asia-Pacific Ocean, and similar incidents would occur every ten years (Lau et al., 2021; Tagaki et al., 2018).

Figure 2

Measured Inundation depths during Typhoon Hato



Note. From “A comparative study of typhoon Hato (2017) and Typhoon Mangkhut (2018) – their impacts on coastal inundation in Macau,” by J. Yang, L. Li, K. Zhao, P. Wang, D. Wang, I. Sou, Z. Yang, J. Hu, X. Tang, K. Mok, & L. Liu, 2019, *Journal of Geophysical Research: Oceans*, 124(12), 9590-9619.

<https://doi.org/10.1029/2019JC015249>

The ravages brought by Hato have delivered a wake-up call for all citizens, urging for a comprehensive disaster management system and civil defense strategies. A complete crisis prevention system, including preventative measures, operational tactics, resolution, assessment, and feedback, is urgently in demand to consolidate tourist confidence (U & So, 2020). In particular, Chan et al. (2018) have presented three pieces of advice for recovery. To begin with, the enhancement and formulation of related laws, regulations, and relevant standards are in dire need. For example, existing typhoon classification standards and specific construction requirements from other regions could be a reference. In addition, to secure the energy supply during disasters, the Macau government is liable for infrastructure construction, boosting the capacity of emergency water and electricity storage. Furthermore, to strengthen public awareness of self-protection against disaster, scientific promotion concerning knowledge of natural disasters, measures overcoming unexpected incidents, and occasional maneuvers are vital.

Psychological Stress

Study reveals that natural disasters and their correlated losses might trigger post-traumatic stress disorder (PTSD), of which the prevalence of PTSD subsequent to the arrival of typhoon Hato was 5.1% in particular (Hall et al., 2019). It is reasonable to recognize that types of media usage could be associated with different extents of trauma exerted on residents. According to the survey conducted in the study, the trend of retrieving news

resources from websites rather than traditional media like television and radio broadcasts is consistent with previous journals, with a majority of 62% reporting that the Internet is the primary medium for acquiring information, versus 30% preferred television as a prioritized approach. It seems logical that citizens intend to search for possible hazards to alleviate their apprehension of uncertainty. Nevertheless, they are eventually inundated with videos featuring drowning victims and negative emotions of residents or the firsthand experience of having encountered house damage, which may consequently increase community stress levels (Hall et al., 2019). More surprisingly, how media interpret news can also influence the sense of stress. The threat of typhoon Hato, overshadowed by its magnitude and rarity, was worsened by the media, signifying a vast degree of hazard. To be brief, multiple correlates are significant precipitates for PTSD, including property loss, life-threatening experiences, the witness of traumatic events, and outage of daily necessities.

Therefore, the Macau government should consider therapeutic strategies for stress relief, as victims suffering from severe mental disorders require multi-faceted assistance, which might aid in developing techniques for catastrophe comfort, reaction, and recovery in the future. However, Shi and Hall (2020) observed that traits of Chinese culture might cause hindrance to the citizens' behavior in seeking help from mental health professionals, as there is a stronger preference to opt for intimate partners, friends, family, or classmates. They further proposed that the supportiveness of professional mental health assistance is a crucial

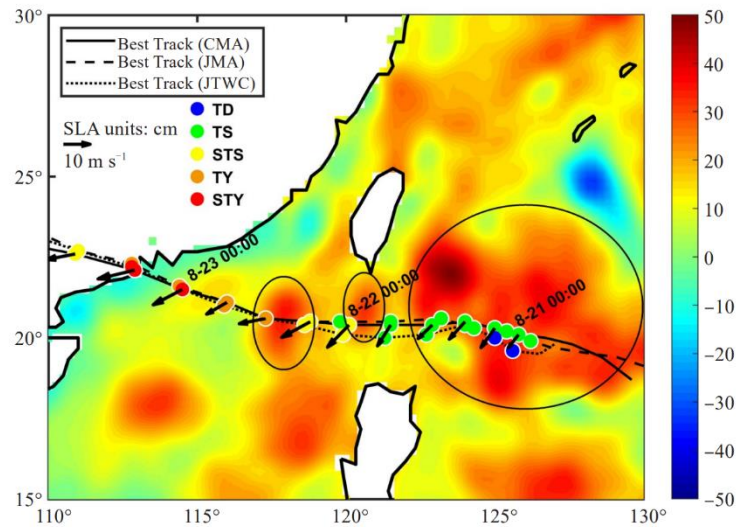
indicator of help-seeking preference. Publicity of the resources, information, services, and treatments of mental assistance services will enhance the probability of outreaching for psychological support (Shi & Hall, 2020). On the other hand, objective information and heroic images of rescue offer positive influences in the post-disaster stage and thus could be protective factors (Hall et al., 2019). Emotional coping and a neutral, objective description of disaster seem probable to ameliorate mental wellness after exposure to disaster-related media.

Forecasting Procedures and Governmental Response

Current research discussed that two meteorological conditions are accountable for the difficulties in forecasting. As the ocean provides heat and moisture for tropical cyclones, Huo et al. (2019) have proposed an experiment verifying that the warm water had a crucial impact on typhoon Hato's rapid intensification process by 20 percent. Another study by Choy et al. (2020) from the Hong Kong Observatory also supported that the convective hot towers that persisted over large areas around Hato had signified rapid intensification. Periodical positioning disturbances are another error source contaminating the precision of prediction. During Hato's landfall, the unstable, dynamic water vapor distribution caused unpredictable GPS signal refraction in the troposphere, with fewer visible satellites available for measurements. With the consistency of rainband patterns matching the worsened performance, it seems evident that the reasons for the degradation of positioning performance can be diminished to a specific range of phenomena (Yu & Liu, 2021).

Figure 3

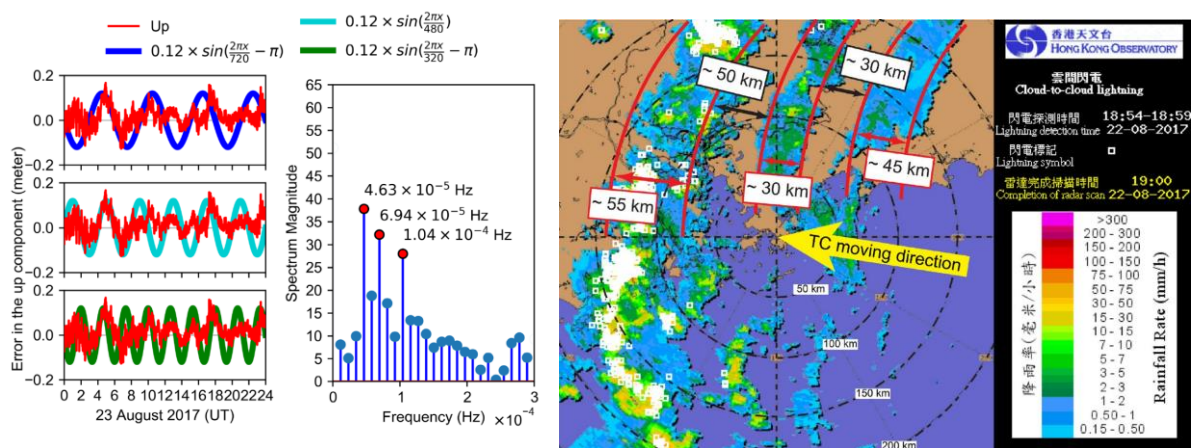
The Distribution of Sea Level Anomaly Pattern on August 19, 2017



Note. From “The effect of warm water and its weak negative feedback on the rapid intensification of typhoon Hato,” by Z. Huo, Y. Duan, & X. Liu, 2020, *Journal of Tropical Meteorology*, 26(4), 402-415, <http://dx.doi.org/10.46267/j.1006-8775.2020.035>

Figure 4

The Positioning Errors on August 23, 2017 and Image of Radar Echoes on August 22, 2017



Note. From “Tropical cyclone-induced periodical positioning disturbances during the 2017 Hato in the Hong Kong region,” S. Yu & Z. Liu, 2021, *GPS Solutions* 25(3): 109, <https://doi.org/10.1007/s10291-021-01112-3>

The irregularities of administrative conduct in government have further deteriorated the consequences caused by the lack of awareness. In the prodromal crisis stage, there are no early warnings of severe catastrophe, and the demonstration of a dilatory response from the Meteorological Bureau of Macau (SMG) has indicated a lack of consciousness for warning (Chan et al., 2018). According to the Commission Against Corruption of Macau (2017), an investigation into the former director of the SMG has revealed the unsound mechanism of discussions in hoisting typhoon signals. Any issuance of typhoon signals is made through phone calls and the Internet at home by personal judgments of the former director. It may seem surprising to learn that ignorance of opinions from frontline forecasters could also be accountable for the exacerbation of the situation.

Hence, several pieces of advice are offered to form better forecasts and assemble informed decisions. Given that most residents felt the delay in the warning signal, the flexibility of sufficient extra time for preparation and adaptation is essential (Tagaki et al., 2019). The government should define areas subject to destruction by natural disasters and concentrate on the response and recovery from the incident provided with the budget (Kim et al., 2019). In order to boost the quality of early forecasts and preventative measures, quantitative evaluation techniques should be adopted to safeguard the city from natural calamities and to maintain long-term social development. For example, current Artificial Intelligence technologies could assist in the clustering of typhoon paths, which could

determine optimal threat mitigation strategy, or utilize prediction models to correct errors 6 hours in advance by machine learning algorithms (Kim et al., 2019; Yu & Liu, 2021).

Conclusion

Tropical Cyclones have always been destructive throughout history (Chow, 2019). To reiterate, this essay has examined several facets of societal impacts created by the 2017 typhoon Hato, namely the economic losses and fatalities, psychological anxiety, and forecasting and governmental responses. It seems undeniable that there are some technical constraints and unfavorable administrative occasions to overcome as yet, but we should be cognizant of our vulnerable society to extreme climate events. Risk management systems, early forecasts, and appropriate public services are essential for recovery from trauma. Further research could continue on this topic to compare with the 2018 typhoon Mangkhut to address the progress of preventative procedures. To accomplish such a goal, we must make a concerted effort to attain cooperation between individuals and the government, and then we will finally be able to overcome similar incidents in the not-so-distant future.

Word Count: 1,618 Words

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